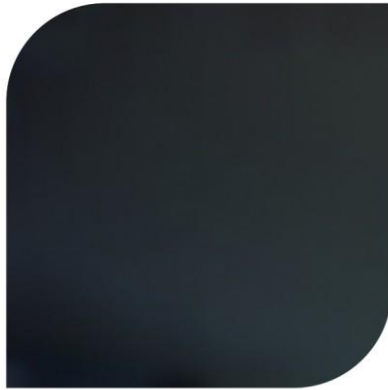
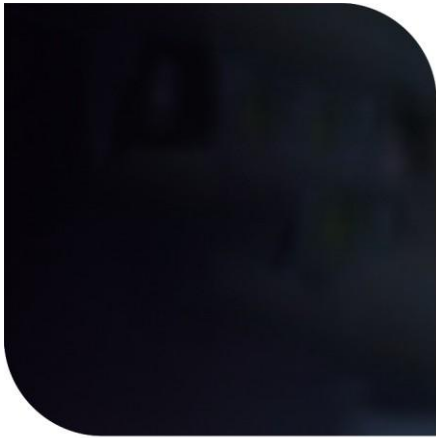


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DOCUMENT QUESTION ANSWERING SYSTEM USING RAG

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Project Title : Document Question Answering

Mentor :

Date of submission :

DOCUMENT QUESTION ANSWERING SYSTEM USING RAG

1. Project Objective

The goal of this project is to design and implement a **Retrieval-Augmented Generation (RAG)** system that can:

- Ingest PDF documents
- Chunk the text for efficient retrieval
- Generate embeddings for semantic understanding
- Store embeddings in a vector database (**in-memory FAISS**)
- Retrieve relevant chunks based on a query
- Generate answers using an LLM
- Display results through an interactive **Streamlit UI**

This project demonstrates understanding of **document processing, embeddings, vector search, retrieval pipelines, and UI integration.**

2. Document Processing

Process:

- PDF documents are uploaded via the Streamlit interface.
- Text is extracted using PyPDF2.
- Document is split into smaller chunks using RecursiveCharacterTextSplitter.

Chunking Strategy:

- Chunk size: 500 characters (balances context and retrieval efficiency)
- Chunk overlap: 50 characters (ensures continuity between chunks for context)

Justification: A chunk size of 500 ensures that the context is sufficient for the LLM to generate accurate answers without overloading memory. An overlap of 50 maintains context across chunks, avoiding loss of information at boundaries.

3. Embeddings

Embedding Model: all-MiniLM-L6-v2 via HuggingFaceEmbeddings

What embeddings represent:

- Embeddings are numerical vectors representing semantic meaning of text.
- Similar texts have closer embeddings in the vector space.

Why embeddings are necessary in RAG:

- Enable semantic search rather than keyword-based search.
- Allow the system to retrieve the most relevant chunks even if query words don't match exactly.

4. Vector Database (FAISS)

Process:

- Each chunk is converted into a vector using embeddings.
- Vectors are stored in in-memory FAISS index (no separate faiss_index folder).
- For a user query, the system retrieves Top-3 most relevant chunks based on vector similarity.
- **Note:** FAISS is used in-memory directly from app.py without creating separate index files.

Enter your question

why do use paracetamol?

Chunk 1

Paracetamol and its side effects

Paracetamol ,[a] or acetaminophen ,[b] is an analgesic and antipyretic agent used to treat fever and mild to moderate pain .[14][15] It is a widely available over-the-counter drug sold generically or under various brand names , including Tylenol and Panadol . Paracetamol relieves pain in both acute mild migraine and episodic tension headache .[16][17] At a standard dose, paracetamol slightly reduces fever,[14][18][19] though it is inferior to ibuprofen in that respect[20] and the benefits of its use for fever are unclear, particularly in the context of

FAISS Score: 0.5930

CrossEncoder Score: 5.0072

Chunk 2

its antipyretic properties, particularly in adults, and thus its benefits are unclear.[14] As a result, it has been described as over-prescribed for this application.[14] In addition, low-quality clinical data indicates that when used for the common cold , paracetamol may relieve a stuffed or runny nose, but not other cold symptoms such as sore throat , malaise , sneezing , or cough .[55] For people in critical care , paracetamol decreases body temperature by only 0.2 –0.3 °C more than control interventions and has no effect on their mortality .[18] It did not change the outcome

FAISS Score: 0.5895

CrossEncoder Score: 4.9773

Chunk 3

paracetamol is due to its quinone metabolite NAPQI and NAC also helps in neutralizing it.[117] Kidney failure is also a possible side effect.[114]

FAISS Score: 0.6130

CrossEncoder Score: 4.3377

5. Retrieval QA Chain

Process:

- Used LangChain's RetrievalQA chain.
- LLM receives retrieved chunks and the user query to generate a final answer.
- Display includes:
 - Source Chunks

6. Streamlit UI

Features Implemented:

- PDF Upload: Users can upload a document.
- Query Input: Text box for user queries.
- Answer Display: Shows the generated answer.
- Retrieved Chunks: Shows source chunks used for generating the answer.
- Loading Spinner: Displays while answer is being generated.



Document Question Answering System

Upload PDF

 Drag and drop files here
Limit 200MB per file • PDF

Browse files

 paracetamol.pdf 365.4KB 

 ml notes.pdf 1.1MB 

Select Document

paracetamol.pdf 

Document split into 30 chunks

Embeddings stored in FAISS

Enter your question

why do use paracetamol?

7. Code Structure

```
RAG_Project/  
|  
├─ app.py           # Main Streamlit app  
├─ helper_functions.py # PDF loading, chunking, embeddings, retrieval  
├─ requirements.txt  # Dependencies  
└─ documents/       # Sample PDF files
```

8. Key Learnings

- Recursive text splitting is crucial for large documents.
- Embeddings allow semantic understanding and retrieval.
- FAISS enables efficient vector search without needing persistent storage.
- LLMs perform better when provided with contextually relevant chunks.
- Streamlit enables simple yet interactive UI for AI applications

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